Cristian Camilo Escandón García



Full Stack Developer

Founder of REMOTEHUBTECH S.A.S. · Creator of RemoteHub

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6 platforms built for Areandina in 2026 **750+** REST endpoints across those platforms **4,000+** automated tests
3 platforms with built-in generative AI

Profile

I am a full-stack developer with more than four years of experience and the founder of **REMOTEHUBTECH S.A.S.**, the company behind **RemoteHub**, a remote access software for Windows and Mac. I work in TypeScript from data model to UI: so far in 2026 I have built six platforms for Fundación Universitaria del Área Andina (Areandina), a Colombian university, two of them already in production. I build NestJS and Express APIs on PostgreSQL, React and Next.js interfaces, desktop apps in Electron and .NET, real-time video over WebRTC, and Docker and AWS deployments, with role-based access control, two-factor authentication, audit logging and automated testing.

My company

REMOTEHUBTECH S.A.S. FOUNDER

Colombian software development company I founded in 2026. Through it I build custom software, provide IT consulting and develop RemoteHub, my remote access product for Windows and Mac.

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Software development ISIC 6201 Planning, analysis, design, programming and testing of software systems.
IT consulting ISIC 6202 Technology consulting and management of IT facilities.
IT services ISIC 6209 Other information technology and computer service activities.

RemoteHub — Your computer, wherever you are.

Remote access software for Windows and Mac: see another computer's screen and control it from your computer or your phone. Built for remote work, businesses and school or university computer labs.

- Remote control from another computer or a phone, with file transfer, chat and multiple monitors.
- Unattended access with a PIN or a daily password, even at the Windows sign-in screen.
- Remote power-on for switched-off computers on a wired network.
- Per-person accounts, invitations, teams, downloadable connection history and a consent-based activity log.
- Two-step verification with apps such as Google Authenticator.
- Time-slot computer lab bookings for schools and universities, with automatic disconnection.
- Screen, files, keyboard and mouse encrypted between both computers; no screen or keystroke recording.

Experience

OCT 2026 – PRESENT

Founder & Full Stack Developer

REMOTEHUBTECH S.A.S. Bogotá D.C., Colombia

- Founded REMOTEHUBTECH S.A.S., a software development and IT services company.
- Built RemoteHub end to end, remote access software for Windows and Mac: real-time video and control over peer-to-peer WebRTC, with DTLS-SRTP encryption and NestJS signaling over Socket.IO.
- Built its three clients: the Next.js web app, the Windows desktop app in C# with .NET 8 (WPF) and an MSI installer, and the native Mac app in Swift.
- Implemented unattended access with a PIN or daily password, remote power-on, file transfer, chat and multiple monitors.
- Implemented per-user accounts with two-step verification, Google and Microsoft sign-in, connection history, time-slot computer lab bookings, and Pro and Business subscriptions billed through Mercado Pago.
- Deployed the API on AWS (ECS Fargate, a load balancer and managed Redis) with infrastructure as code in CloudFormation and continuous deployment from GitHub Actions.

Tech stack: TypeScript, NestJS, Next.js, React, WebRTC, Socket.IO, PostgreSQL, Prisma, Redis, C# / .NET 8 (WPF), Swift, Tailwind CSS, Docker, AWS, GitHub Actions

MAR 2026 – PRESENT

Full Stack Developer (Professional)

Fundación Universitaria del Área Andina (Areandina) Bogotá D.C., Colombia

- Build complete web platforms for several university departments, from data model and API to UI, deployment and documentation, using TypeScript, React, Next.js, NestJS, PostgreSQL and Docker.
- Shipped Acopia, which replaced manual tracking of equipment, supplies and maintenance in the health and sports laboratories: a kardex-style stock ledger with lots and FEFO issuing, signed requests, CMMS maintenance, two-factor authentication and Docker deployment on an institutional server.

- Developed PMO Enterprise for the Administrative and Financial Vice-Rectorate's project office, where each department manages projects, budgets, costs, goals and committees, with earned value (EVM) metrics and an AI report generator.
- Built the campus touchscreen kiosk platform (Quioscos interactivos): an Electron desktop app, a Next.js admin panel and a NestJS API with real-time Socket.IO monitoring and remote commands. The API and panel are already deployed to the cloud.
- Building two platforms not yet in production: PIS, the Institutional Sustainability Platform (a Next.js portal and 6 NestJS microservices with Keycloak, Kong, RabbitMQ and Airflow, a tamper-evident audit log and 2,500+ automated tests), and GPS, a document workflow manager with approval routing, OTP-based e-signatures and an AI assistant that drafts documents on official templates.
- Built PlanMejora, a working prototype with a Next.js portal and five NestJS microservices that replaces the Google Apps Script app for academic improvement plans, including an ETL from the legacy Google Sheet.

Tech stack: TypeScript, React, Next.js, NestJS, Node.js, Express, PostgreSQL, Prisma, Docker, Electron, Socket.IO, Keycloak, RabbitMQ, Redis, GitHub Actions, NVIDIA NIM

`FEB 2022 – MAR 2026`

Freelance Full Stack Developer

Freelance Bogotá D.C., Colombia

- Built custom software and websites for clients.
- Worked with PHP and MySQL, Python (Django and Flask), JavaScript and React, and Node.js with TypeScript.

Tech stack: PHP, MySQL, Python, Django, Flask, JavaScript, React, Node.js, TypeScript

Featured projects

`Private codebase · owned by Areandina`

Acopia — Full-stack inventory and maintenance management (CMMS) platform for labs

`AREANDINA · LABORATORIOS DE SALUD Y DEPORTE · 2026 · In production on an institutional server`

Replaced manual tracking of equipment, supplies and maintenance in the university's laboratories.

121 REST API endpoints **17** web app screens **242** automated test cases

- Designed a 6-stage request workflow engine with area approval, per-stage touch signatures, stock deduction based on actual consumption and a PDF delivery record.
- Built the CMMS module: work orders, a maintenance and calibration calendar, daily email alerts, satisfaction surveys and MTBF, MTTR and availability KPIs.
- Implemented RBAC with a per-company configurable permission matrix enforced by guards on 51 endpoints, plus mandatory two-factor authentication (2FA/TOTP) and a read-only guest mode.
- Moved the deployment from Vercel, Render and Supabase to Docker Compose on an institutional server, with nginx proxying the API, an idempotent installer, health checks and backups.

Tech stack: TypeScript, React, Vite, NestJS, Prisma, PostgreSQL, JWT + TOTP, Docker Compose, nginx, Jest, Vitest, Google Drive API, Nodemailer

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PMO Enterprise — Full-stack platform for projects, budgets and committees

`AREANDINA · OFICINA DE PROYECTOS DE LA VICERRECTORÍA ADMINISTRATIVA Y FINANCIERA · 2026 · In production on a university server`

Projects, budgets, goals and committees for a vice-rectorate, plus AI reports with no made-up figures.

19 business modules in the API **121** REST endpoints across 21 controllers **259** test cases (67 unit and 192 end-to-end)

- Built the full platform (NestJS, React and PostgreSQL) where each department records projects, budget caps, costs, goals and committees.
- Designed an LLM document generator on NVIDIA models in which the model never writes figures: code fills in the placeholders and a DOM auditor rejects any unbacked digit.
- Built budget control with per-area caps and transactional row locks, COP/USD conversion using Colombia's official TRM exchange rate from datos.gov.co, and earned value (EVM) indicators.
- Automated Docker Compose deployment with a Python packager and Bash installers that back up first, check migrations with SQL and can roll back to the previous version.

Tech stack: TypeScript, React, Vite, Tailwind CSS, TanStack Query, AG Grid, NestJS, Prisma, PostgreSQL, MinIO, NVIDIA NIM (Nemotron), Zod, Google Identity Services, Docker Compose, nginx, Jest, Supertest

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Quioscos interactivos — Touchscreen kiosk network: API, web admin panel and desktop app

`AREANDINA · 2026 · Deployed: API and web panel in the cloud`

Built to manage, customize and monitor the campus touchscreen kiosks in real time.

3 + 1 apps (API, admin panel and kiosk) plus a shared package **176** REST endpoints **25** admin panel screens

- Developed the Electron and React kiosk app: locked-down kiosk mode, automatic device registration, 9 interactive modules (maps, games, recorder, whiteboard and 3D graph) and NSIS and MSI installers.
- Added real-time fleet monitoring with Socket.IO: hardware heartbeats, automatic offline and resource alerts, remote commands (restart, shutdown and cache clearing) and a screenshot-based live view.
- Implemented the NestJS API on PostgreSQL and Prisma, with JWT and rotating, revocable refresh tokens, role-based access control, change auditing and per-company isolation.
- Designed a Company → Group → Kiosk configuration cascade that customizes each device from a single catalog and applies changes in real time.

Tech stack: TypeScript, Electron, React, Next.js, NestJS, Socket.IO, PostgreSQL, Prisma, Tailwind CSS, shadcn/ui, TanStack Query, React Flow, Three.js, Docker, Render, Vercel, Neon

PIS Institutional Sustainability Platform (microservices monorepo)

`AREANDINA · SUBDIRECCIÓN NACIONAL AMBIENTAL Y DE SOSTENIBILIDAD · 2026`
`· In development; runs fully in a local environment, not yet in production`

Sustainability indicators captured once, validated and approved, feeding AI-drafted reports that use only approved figures, with human review.

6 NestJS microservices **2,500+** automated tests (Jest, Vitest, Playwright and pytest)
10 GitHub Actions workflows (CI, quality, security and E2E)

- Built the platform: a Next.js portal with 15 screens and 6 NestJS microservices on PostgreSQL, RabbitMQ, Redis and Kong, in a pnpm monorepo.
- Integrated Keycloak 26 as the sole token issuer: OIDC with PKCE in the portal, JWKS validation in 5 services, and RBAC plus area-based ABAC.
- Implemented a tamper-evident audit log (SHA-256 hash chain) in PostgreSQL: partitioned table, sealing triggers, advisory locks and scheduled integrity checks.
- Adapted and integrated the AI report generator with 3 switchable providers (NVIDIA NIM, Azure OpenAI and Gemini), PII and prompt-injection guardrails, and four-eyes human approval.

Tech stack: TypeScript, Next.js, NestJS, Prisma, PostgreSQL, RabbitMQ, Redis, Keycloak, Kong, Apache Airflow, Python, Docker Compose, GitHub Actions, Prometheus, Grafana, Jest, Playwright, NVIDIA NIM, Azure OpenAI, Gemini

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GPS Document workflow manager with an AI assistant

`AREANDINA · SUBDIRECCIÓN NACIONAL DE PROCESOS · 2026 · In active development, not yet in production`

From draft to signature: document management with an approval workflow, e-signatures and an AI assistant.

100 REST API endpoints **19** Prisma data models **20** design-system React components

- Implemented an AI assistant (NVIDIA Nemotron) that retrieves passages from institutional PDF, Word and Excel files, streams its answers and drafts documents on official templates.
- Implemented OTP-based electronic signatures with an audit seal (time, IP and device), five-role access control and a state machine for the document lifecycle.
- Designed a custom design system: 20 React components with .d.ts contracts, CSS tokens with dark mode and 14 visual guidelines used by the web app.
- Set up the Docker Compose environment with health checks and migrations on startup, and integrated LibreOffice and Poppler for page-by-page attachment previews.

Tech stack: TypeScript, JavaScript, React, Vite, Node.js, Express, Prisma, PostgreSQL, NVIDIA Nemotron, OpenAI SDK, Server-Sent Events, Docker Compose, LibreOffice, Poppler, ExcelJS

PlanMejora Improvement plans platform (microservices monorepo)

`AREANDINA · PLANES DE MEJORAMIENTO ACADÉMICO · 2026 · Working prototype tested locally, not yet in production`

Replaces a Google Apps Script app so academic programs can manage their improvement plans with roles, traceability and auditing.

5 NestJS microservices **80** HTTP endpoints **1,100+** test cases

- Built a Next.js portal and five NestJS microservices behind Kong to replace the Google Apps Script app used to manage academic improvement plans.
- Built an ETL pipeline (TypeScript CLI plus an import endpoint) that validates the 83-column legacy Google Sheet and loads it with a dry run and idempotent upserts.
- Centralized Keycloak token validation, MFA-aware RBAC/ABAC guards, RFC 7807 errors and Redis-backed idempotency in a shared NestJS library used by all five microservices.
- Automated monorepo verification: 154 test files (Jest, Vitest and Playwright), line coverage above 82% in nine packages and an architecture-conformance checker.

Tech stack: TypeScript, Next.js, NestJS, Prisma, PostgreSQL, Keycloak, Kong, RabbitMQ, Redis, MinIO, Docker Compose, Prometheus, Grafana, Jest, Vitest, Playwright

Skills & technologies

Programming languages: TypeScript, JavaScript, SQL, Python, C#, Swift, HTML/CSS, Bash/Shell
Frontend & desktop: React, Next.js, Vite, Tailwind CSS, TanStack Query, Electron, .NET (WPF), Design systems
Backend: Node.js, NestJS, Express, Prisma, API REST, Socket.IO, Swagger / OpenAPI
Data & messaging: PostgreSQL, Redis, RabbitMQ, MinIO (S3), Apache Airflow
Security: JWT, RBAC, MFA / 2FA (TOTP, OTP), OpenID Connect, Keycloak
DevOps: Docker, Docker Compose, Git / GitHub, GitHub Actions, AWS (ECS Fargate, CloudFormation), nginx, Prometheus y Grafana
Testing: Jest, Vitest, Playwright, Supertest
Generative AI: NVIDIA NIM (Nemotron), Azure OpenAI, Google Gemini, LLM guardrails
Architecture: Microservices, Event-driven architecture, API Gateway (Kong), Real time (WebSockets / SSE), WebRTC (P2P video)
Also work with: PHP, MySQL, Django, Flask

Languages

Spanish: Native **English:** Intermediate (B1–B2)